

Summary: Opioids like fentanyl are very good but have some nasty complications like addiction. Used computational chem/bio tools to study derivatives of fentanyl that bind to another receptor (NOP) alongside primary receptor (MOR), which has been shown in a few other drugs to reduce side-effects. First used random forest regressor/statistical analysis on compounds from ChEMBL (large drug database) to weigh "important features" (figures 1-3), then applied that on another database called ZINC22 to get molecule attachments for norfentanyl (figures 4-7). Filtered for toxicity (ADMET), then did physics-based docking for Gibbs Free energy/enthalpy and bond entropy calculations for overall affinity. Then did Molecular dynamics for RMSD/standard deviation calculations from target proteins for bond stability. Double checked binding using some other software called PyMol (figures 8-10).

Extended paper portions:

The opioid epidemic highlights the urgent need for analgesics that maintain efficacy while minimizing severe side effects like addiction and respiratory depression. This computational chemistry project focuses on designing novel fentanyl derivatives that co-target the Mu-opioid receptor (MOR) and the Nociceptin receptor (NOP). By designing dual-active ligands, this research aims to leverage a biochemical mechanism that has been shown to successfully reduce adverse complications in other pharmaceutical compounds.

To identify the key physicochemical properties driving this dual-selectivity, a Random Forest regressor and statistical analysis were applied to large-scale compound data from the ChEMBL database. The machine learning model successfully distinguished MOR-exclusive from NOP-exclusive ligands with a 94.2% cross-validation test accuracy. From there, Principal Component Analysis (PCA) and normalized property radar charts were utilized to map the chemical space and pinpoint the precise, overlapping characteristics required for a molecule to successfully bind to both receptors.

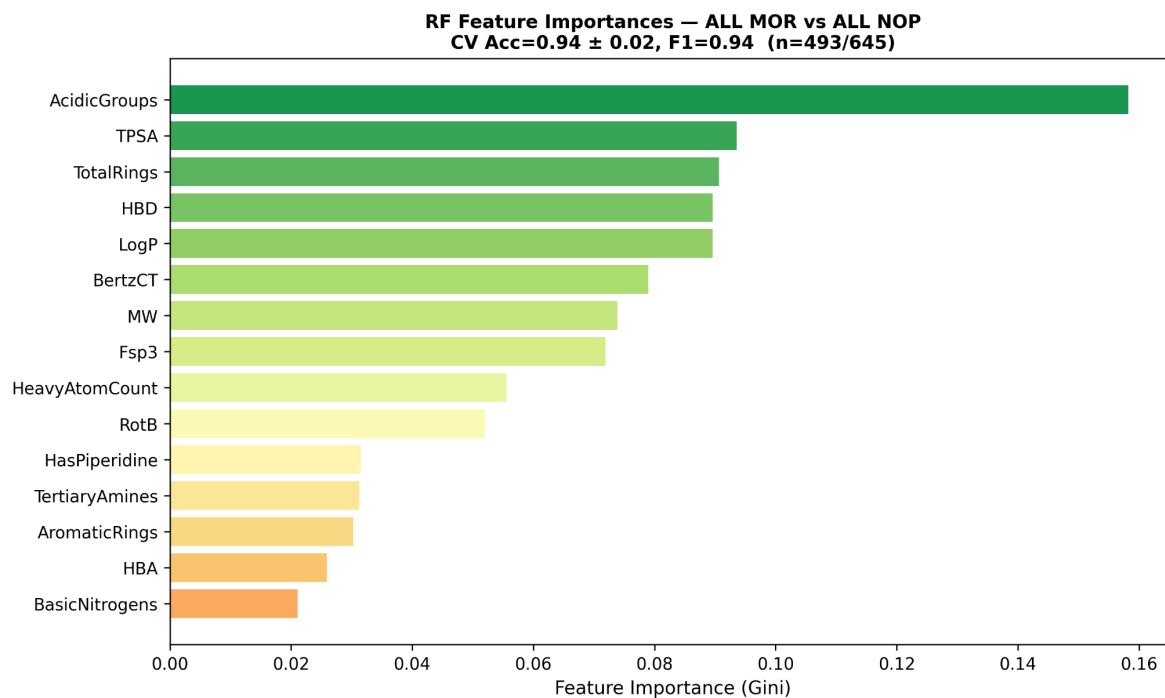


Fig 1: Feature Importance

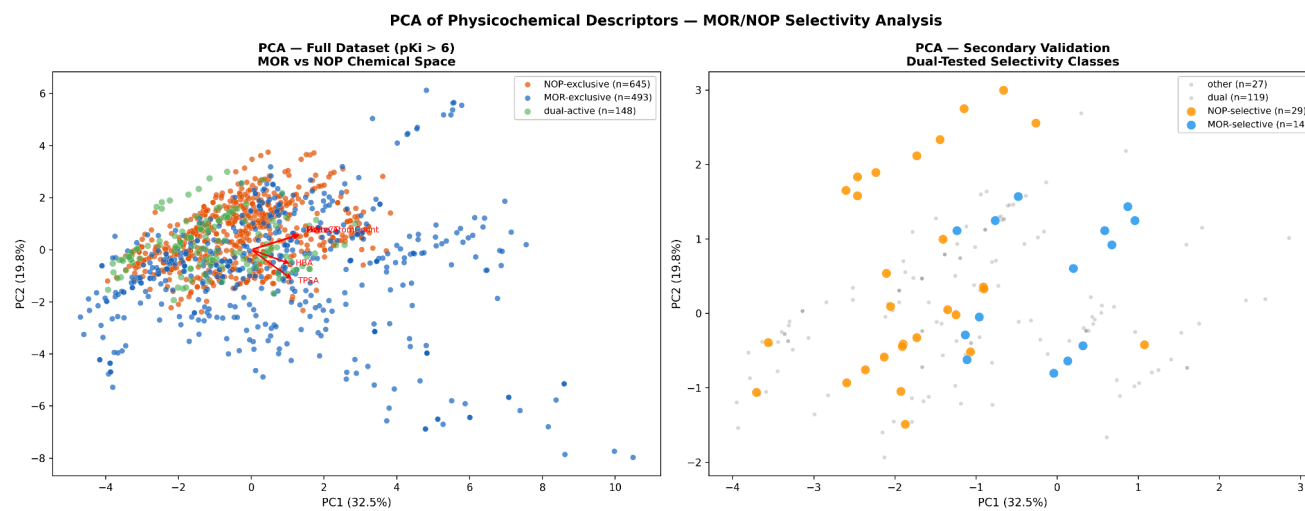


Fig 2: PCA Analysis - Features Driving Receptor Selectivity

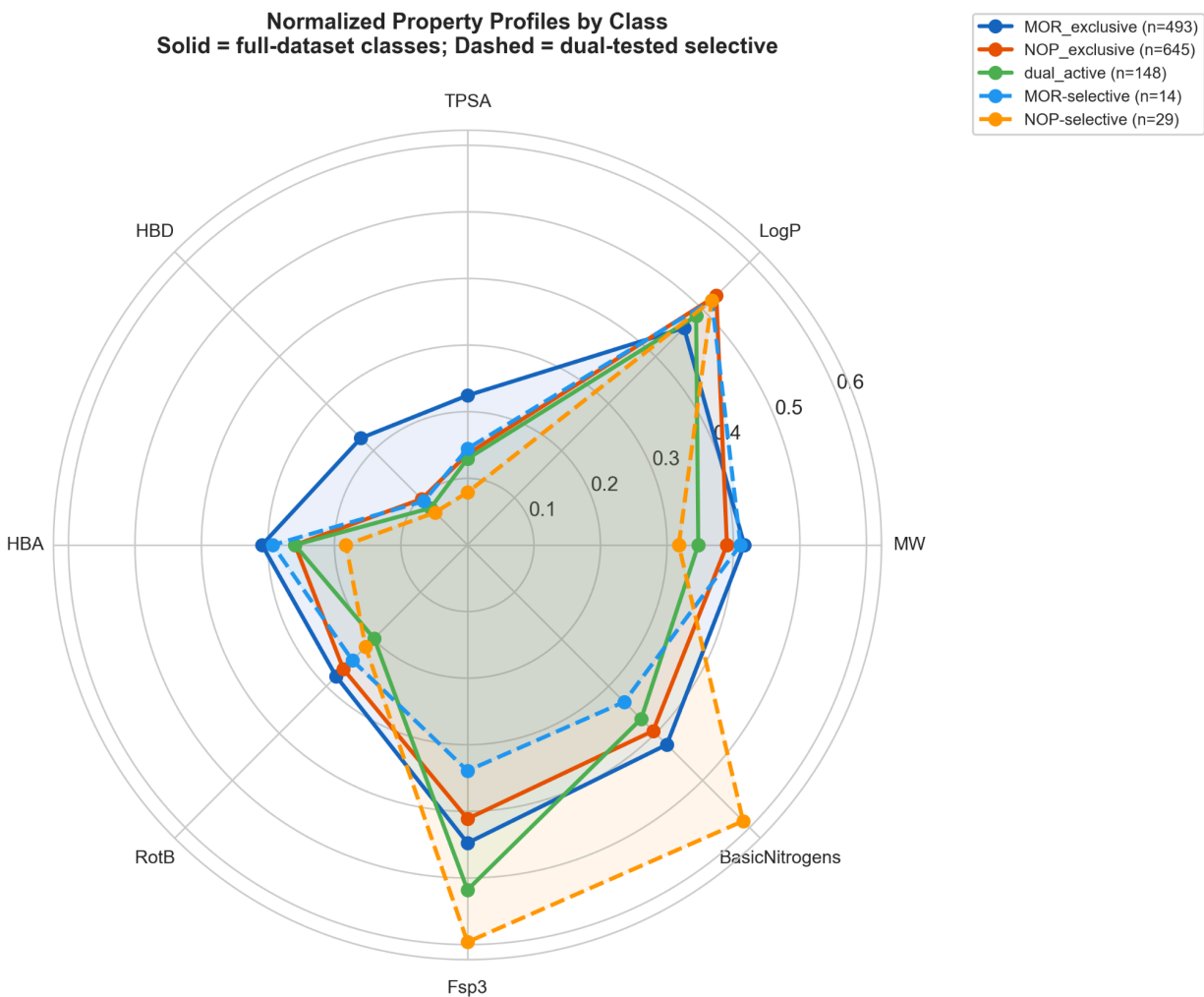


Fig 3: Property Radar Chart - Overlapping Characteristics

Building on the statistical model, the project utilized the ZINC22 database to isolate specific molecular fragments that heavily favor either NOP or MOR binding. This fragment enrichment analysis allowed for the extraction of highly effective functional groups. These functional groups were then strategically attached to a norfentanyl backbone, resulting in several viable structural scaffolds that exhibit the exact chemical profiles necessary for dual activation.

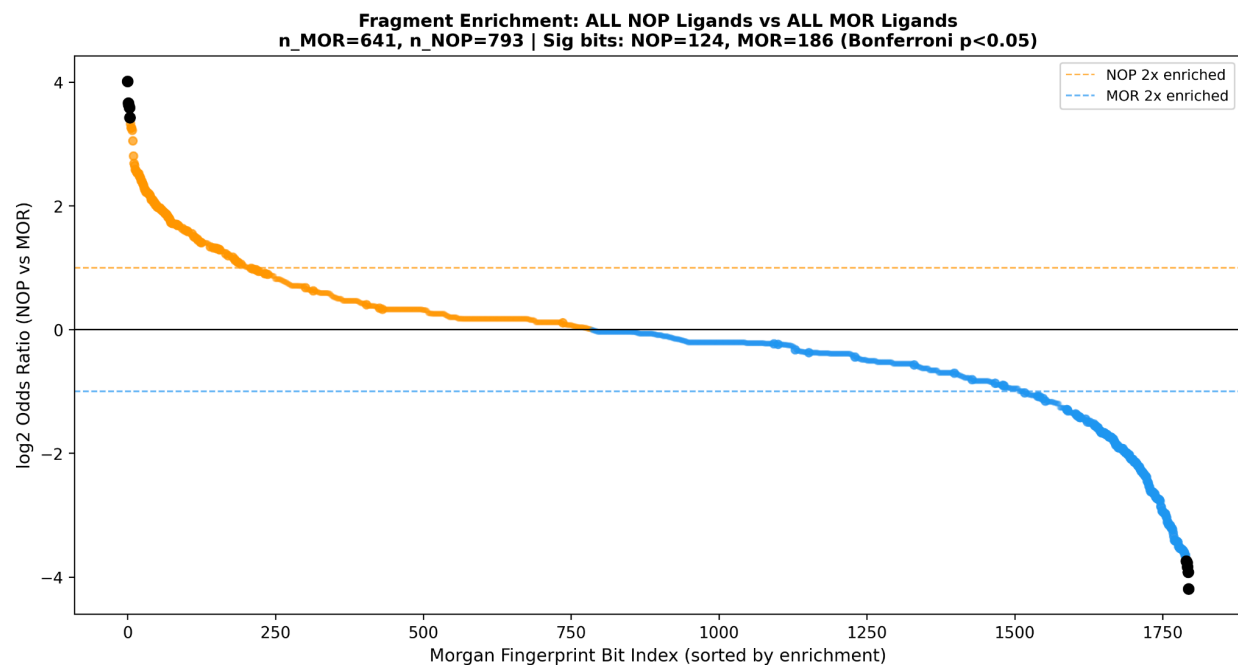


Fig 4: Fragment enrichment volcano plot-statistical distribution of molecular bits enriched in

MOR vs NOP Ligands

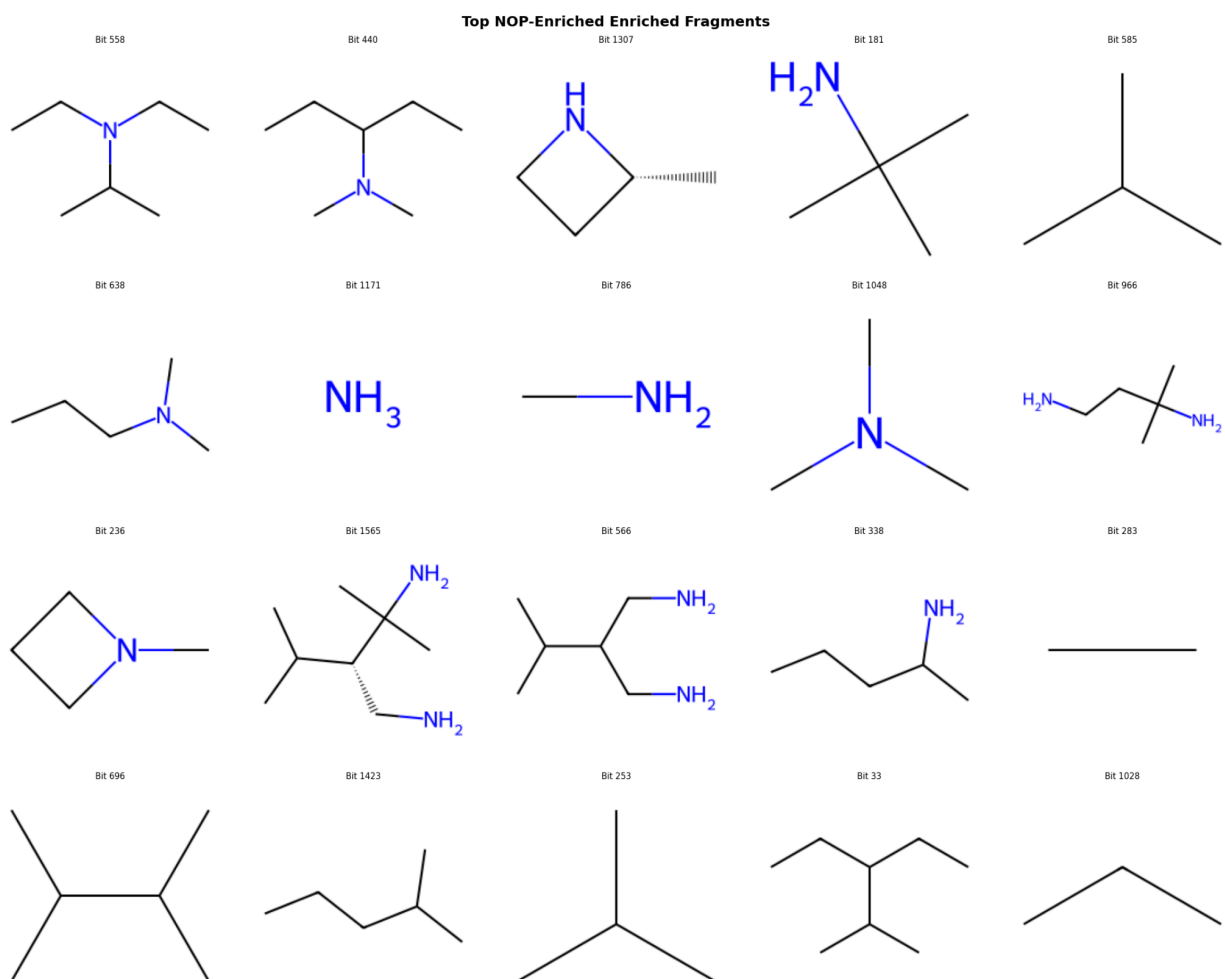


Fig 5: Top NOP-Enriched Fragments - Statistically significant contributors to NOP affinity

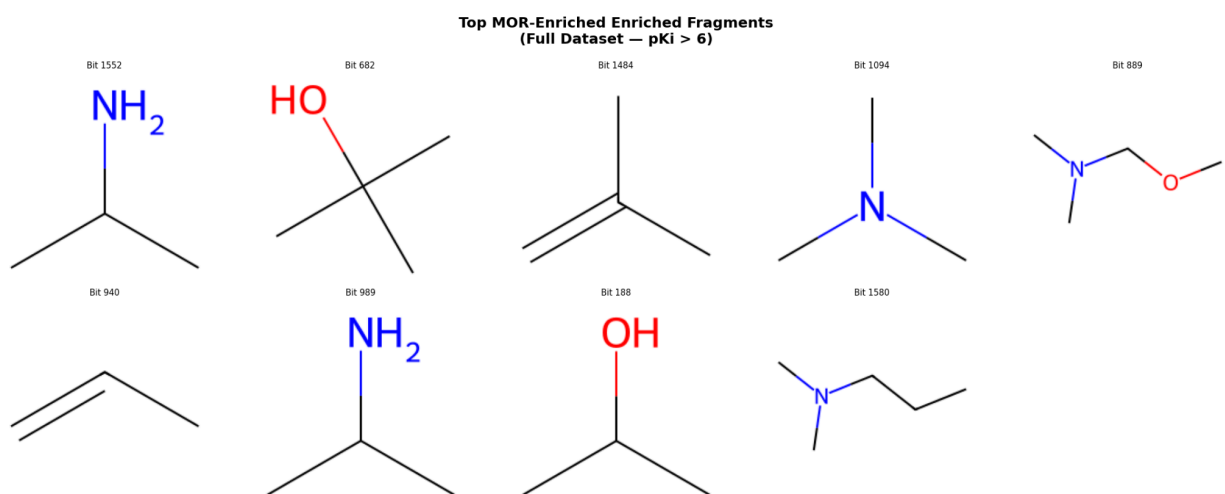


Fig 6: Top MOR-Enriched Fragments - Statistically significant contributors to MOR affinity

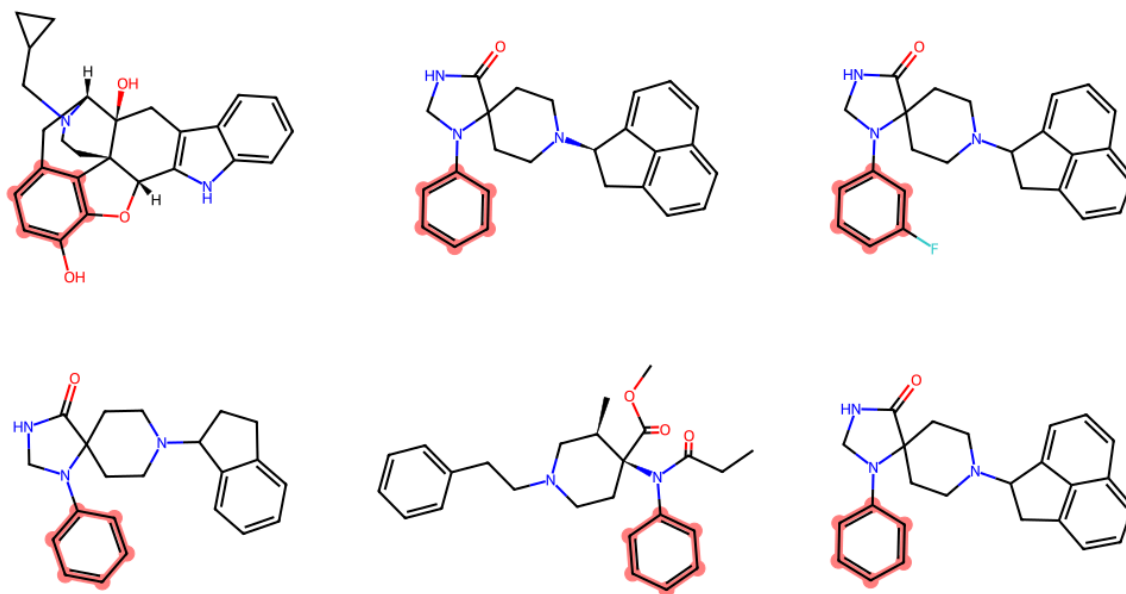


Fig 7: Common Lead Scaffolds – Representative chemical templates identified for dual-active compounds, utilized for the design of novel norfentanyl derivatives.

Finally, the newly generated compounds were filtered for toxicity using ADMET profiling before undergoing rigorous physics-based simulations to validate real-world viability. AutoDock Vina was utilized to calculate Gibbs Free Energy (ΔG), enthalpy, and bond entropy to determine the overall binding affinity of the compounds within the target protein pockets. Following the initial docking, Molecular Dynamics (MD) simulations were run to calculate the Root Mean Square Deviation (RMSD) to ensure the stability of the ligand-receptor bonds over time, with final 3D structural verification and hydrogen bond analysis completed using PyMOL.

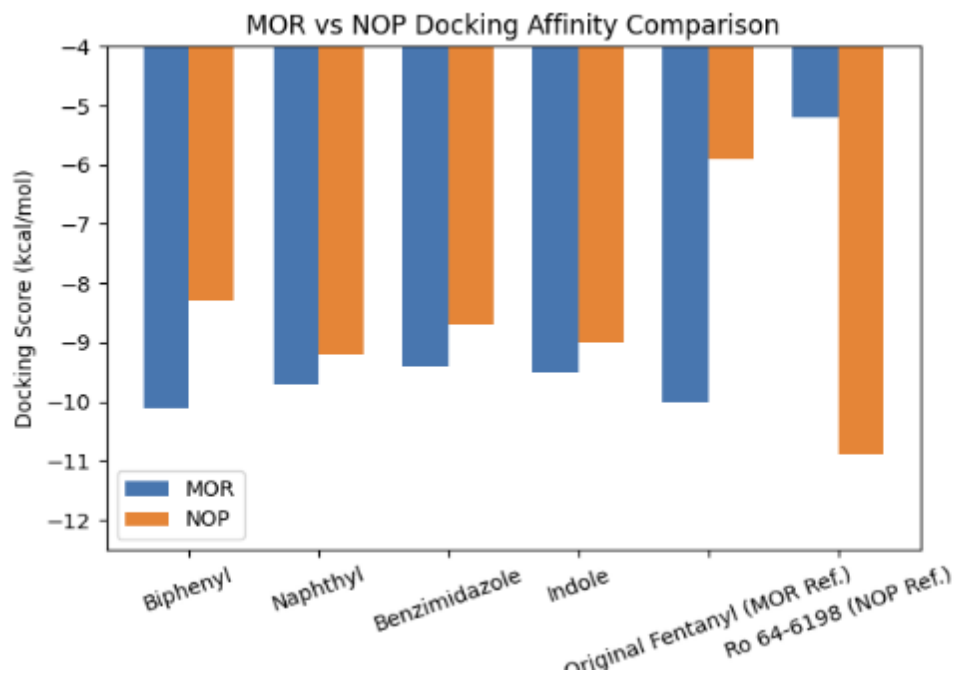


Figure 8: AutoDock Vina Binding Affinity – Calculated Gibbs Free Energy (ΔG) scores representing the thermodynamic favorability of top candidates in target protein pockets.

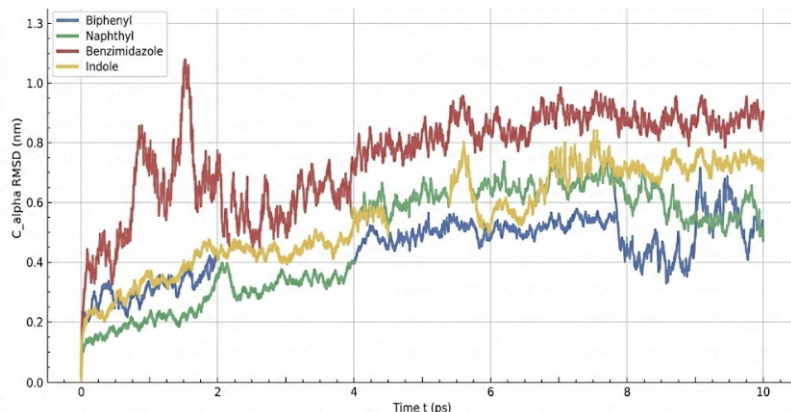
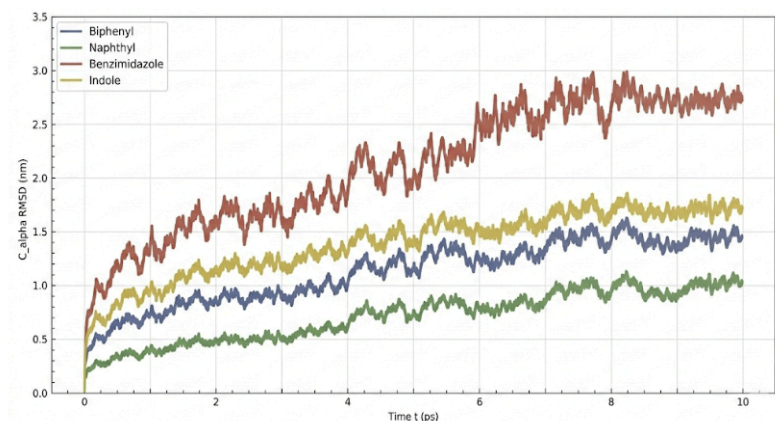


Figure 9: RMSD Stability Trajectories – Root Mean Square Deviation calculations from Molecular Dynamics (MD) simulations to assess the structural stability of ligand-receptor complexes over time.

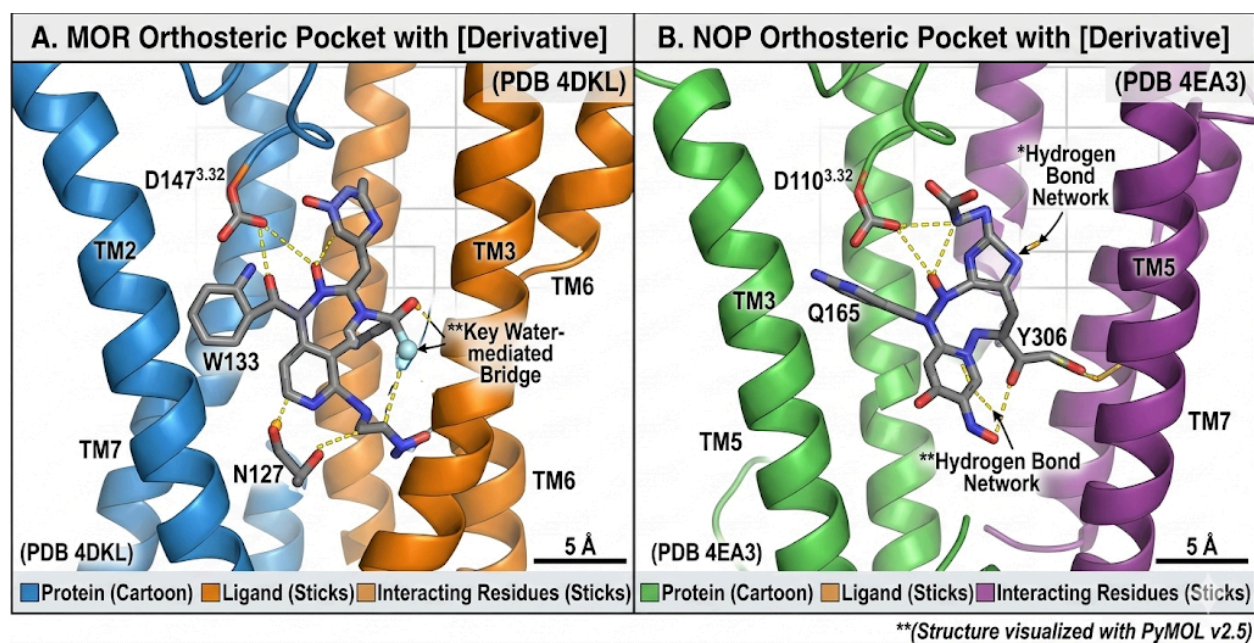


Figure 10: 3D Binding Pocket Visualizations – High-resolution PyMOL renderings highlighting specific hydrogen bonding and hydrophobic interactions between the derivative and the receptor.